General Computer-Style Notation

Up to this point, we have been writing equations and expressions using standard mathematical notation, including symbols such as +, −, ×, ÷, and √. When mathematical expressions are entered into a computer program, different symbols and notation are often needed.

For example, an asterisk (*) is commonly used for multiplication and a slash (/) is commonly used for division. Mathematical expressions must also be written on a single line, so parentheses are often needed to show which operations should be performed together.

The following notation will be used throughout this course:

Mathematical Operation	Written Notation	Computer Notation
Addition	$a + b$	a + b
Subtraction	$a - b$	a - b
Multiplication	$a \times b$	a * b
Division	$a \div b$	a / b
Exponentiation	a^b	a^b
Square root	$\sqrt{a}$	sqrt(a)
Less than	$a < b$	a < b
Less than or equal to	$a \leq b$	a <= b
Greater than	$a > b$	a > b
Greater than or equal to	$a \geq b$	a >= b

Note: The exact notation used may vary depending on the programming language. The notation shown above is the notation we will use in this course.

Let's look at some examples of writing expressions in computer notation.

Written Expression	Computer Notation	Notes
$5 \div 3$	5 / 3	The “slash” is commonly used to indicate division or fractions
7×9	7 * 9	The asterisk is commonly used to indicate multiplication.
4x	4 * x	There must always be an asterisk where multiplication is performed.
$\frac{1}{2}x + 5$	(1 / 2) * x + 5	The parentheses around 1 / 2 make it clear that only the 2 is in the denominator.
$\frac{1}{2x} + 5$	1 / (2 * x) + 5	The parentheses make it clear that both 2 and x are in the denominator.
$\frac{1}{2x + 5}$	1 / (2 * x + 5)	The parentheses make it clear that the 2 * x + 5 is in the denominator.
$\frac{x + 2}{x^2 + 5x + 6}$	(x + 2) / (x^2 + 5 * x + 6)	Parentheses are used to clearly indicate the numerator and denominator.
$\frac{\frac{x + 3}{5}}{\frac{x - 4}{2}}$	((x + 3) / 5) / ((x - 4) / 2)	Use parentheses to clearly indicate the numerator and denominator of each fraction. Using extra spaces is helpful, too.
x^2	x^2	The caret, ^, usually Shift + 6 on the keyboard, is used in this course to indicate exponents.
$x^{\frac{2}{3}}$	x^(2 / 3)	Parentheses are needed to show that 2 / 3 is the entire exponent. Without them, x^2 / 3 means x^2 divided by 3.
$x^2y^3z^4$	x^2 * y^3 * z^4	There must always be an asterisk where multiplication is performed.
$b^2 - 4ac$	b^2 – 4 * a * c	There must always be an asterisk where multiplication is performed.
$\sqrt{2y}$	sqrt(2*y)	The abbreviation sqrt stands for square root, and the parentheses make it clear that both the 2 and the y are inside the radical.

Examples

Convert the expressions from written notation to computer notation.

$a \div b$

$5a^3$

$\sqrt{5x+1}$

$3x^2 - 4x + 7$

$$\frac{x^2 + x + 3}{x^2 + x}$$

$$\frac{x}{4 + \sqrt{x}}$$

$(-32y^{10})^{\frac{1}{5}}$

Convert the expressions from computer notation to written notation.

2 * x * y – z^3

(2 * x^2) / 3

4 * x + x * sqrt(y)

(x + 3)/ x – (x - 1)/(x + 3)

2.4 Distinguish between constants and variables

Variables

In a computer program, data is stored in variables. You are already familiar with the term variable from algebra, where a variable can represent an unknown value. For example, in the Pythagorean Theorem, $a^2 + b^2 = c^2$, the letters a, b, and c are variables. In this formula, a and b represent the lengths of the two shorter sides of a right triangle, while c represents the length of the hypotenuse.

However, variables in programming are somewhat different from variables in mathematics. In programming, a variable is a named storage location in a computer's memory where data can be stored. A variable can hold data entered by the user, the result of a calculation, or another piece of information needed by the program.

The data stored in a variable is called its value. This value can change while a program is running. In this way, variables allow a computer program to "remember" and work with data.

For example, in a computer program, you may see a line of code such as this one:

```
area = pi * r * r
```

This may look like a mathematical equation, but it is not. It is an assignment statement - an instruction that tells the computer to calculate a value and store the result.

The general form of an assignment statement is:

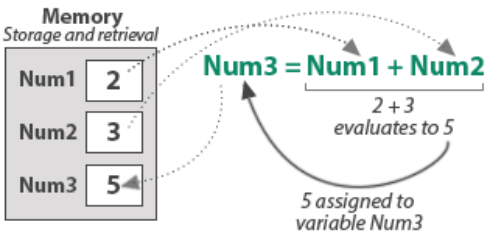
```
variable = expression
```

There is a single variable on the left side of the equals sign and an expression on the right. When the computer processes the statement, it:

- Evaluates the expression on the right.
- Stores the resulting value in the variable on the left.

Therefore, in programming, the equals sign in an assignment statement does not mean "is equal to" in the same way it does in mathematics. Instead, it means "assign the value on the right to the variable on the left."

This process is illustrated below using the assignment statement Num3 = Num1 + Num2.



Now consider the following assignment statement:

x = x + 1

If we interpret this as an algebraic equation, it makes no sense - no number can be equal to itself plus one. However, this is an assignment statement, not an equation.

The computer processes the statement from right to left:

- Retrieve the current value of x.
- Add 1 to that value.
- Store the result back in x.

For example, if x contains the value 0 before the assignment statement is executed, x will contain the value 1 afterward.

Constants

A constant is like a variable because it is used to represent a value in a computer program. The important difference is that the value of a constant does not change while the program is running.

For example, consider the calculation:

area = pi * r * r

In this statement, r may be a variable because its value can change depending on the radius of the circle. The value of pi, however, does not change. Therefore, pi can be defined as a constant.

For example: pi = 3.14159

Once pi has been defined as a constant, the program can use its value whenever it is needed without changing it.

Constants are useful for values that should remain the same throughout a program. Examples might include a tax rate, the number of days in a week, or a fixed conversion factor.

The main difference between a variable and a constant is:

	Variable	Constant
Value	Can change while the program is running	Does not change while the program is running
Example	radius = 5	PI = 3.14159
Purpose	Stores data that may change	Stores a fixed value

For example, suppose a program calculates the area of different circles. The value of the radius may change each time a new circle is entered, but the value of pi remains the same.

Examples

Determine whether the statement is a proper assignment statement. If it is not, explain why.

	Proper	Reason
X = 2		
A + 3 * 2		
9 = Y		
D = C – 2		
4 * A = 7 * B		
C = C + 1		